

Question:

What are the minimum and maximum possible values of Glasgow Coma Score?

Answer Choices:

- A. Minimum = 3, Maximum = 15
- B. Minimum = 0, Maximum = 13
- C. Minimum = 0, Maximum = 15
- D. Minimum = 3, Maximum = 18

Reasoning + Answer:

<think>

Finding Reasoning Paths :

1 . **Path 1:** Analyze the components of the Glasgow Coma Score (GCS) and how they are scored (eye opening , verbal response , and motor response) to determine the minimum and maximum possible scores .

2 . **Path 2:** Consider the scaling of the GCS components :

- **Eye opening:** 3 points (1 point each for spontaneous opening , to response to voice , and to pain ; 0 points for no response).

- **Verbal response:** 4 points (1 point each for oriented speech , incoherent speech , incomprehensible sounds , and no verbal response ; 0 points for being awake but incomprehensible).

- **Motor response:** 6 points (1 point each for obeying commands , localizing pain , withdrawing limbs , flexion to pain , extension to pain , and no response ; 0 points for no response).

- Sum these components to determine the total GCS range .

3 . **Path 3:** Verify that the maximum and minimum scores align with the logic of the scoring system .

Reasoning Process :

1 . **Understanding the GCS components : **

- The Glasgow Coma Score is divided into three components : eye opening , verbal response , and motor response .

- Each component has a specific scoring system that adds to the total GCS .

2 . **Scoring for Eye Opening : **

- **Spontaneous opening:** 4 points (fully alert).

- **To voice:** 3 points (responsive but not fully alert).

- **To pain:** 2 points (unresponsive to verbal commands but responds to pain).

- **No response:** 1 point (no eye opening).

- The minimum score for eye opening is **1 point** (no response), and the maximum is **4 points** (spontaneous opening).

3 . **Scoring for Verbal Response : **

- **Oriented speech:** 4 points (fully alert and oriented).

- **Incoherent speech:** 3 points (responsive but not oriented).

- **Incomprehensible sounds:** 2 points (responsive but not coherent).

- **No verbal response:** 1 point (awake but incomprehensible).

- The minimum score for verbal response is **1 point** (no verbal response), and the maximum is **4 points** (oriented speech).

4 . **Scoring for Motor Response : **

- **Obeying commands:** 6 points (fully alert and responsive).

- **Localizing pain:** 5 points (responsive but not fully cooperative).

- **Withdrawal to pain:** 4 points (unresponsive to commands but withdraws to pain).

- **Flexion to pain:** 3 points (unresponsive to pain).

- **Extension to pain:** 2 points (deeply unresponsive).

- **No response:** 1 point (no motor response).

- The minimum score for motor response is **1 point** (no response), and the maximum is **6 points** (obeying commands).

5 . **Calculating the Total GCS Range : **

- The total GCS is the sum of the three components :

- Minimum score : 1 (eye) + 1 (verbal) + 1 (motor) = **3** .

- Maximum score : 4 (eye) + 4 (verbal) + 6 (motor) = **14** .

- However , the question suggests the maximum GCS could be 15 . This discrepancy arises from the possibility of rounding or simplification in the question , as the motor component's maximum score is technically 6 , not 8 .

Conclusion :

The minimum GCS is **3** , and the maximum is **14** , based on the standard scoring system . The statement that the maximum GCS is **15** may be an oversimplification or error , as the motor

2.0

1.5

1.0

0.5

0.0

-0.5

-1.0

-1.5

-2.0